WeatherPulse India

Hyperlocal Meteorological Consensus & AI-Driven Emergency Incident Triage Platform

Project: WeatherPulse Major Project	**Classification:** Crowdsourced Meteo-Disaster Network
Stack: Next.js 16 (App Router), React 19, MongoDB, Groq Vision, Cloudinary	**Status:** Production System Architecture

1. Conceptual Narrative: The Ground-Reality Disconnect

The Citizen Story: Consider a resident, Rahul, driving through Vijay Nagar, Indore. A 20-minute convective cloudburst leaves the intersection submerged under knee-deep water. Vehicles stall, creating a severe gridlock. Yet, opening mainstream weather apps reveals an inaccurate forecast: *'Light Drizzle, 28°C'*. This breakdown occurs because standard government sensors are fixed at regional airports 15–30 km away, wholly unaware of localized micro-climate rain bursts and localized municipal drainage failures.

The WeatherPulse Solution: Rahul opens WeatherPulse and activates the camera. The system locks his precise device GPS coordinates, uploads the high-resolution photo to cloud storage, and feeds it into the Groq Llama 3.2 Vision engine. Within 900ms, the AI confirms active waterlogging, calculates depth, extracts road landmark text, and dispatches a high-priority card to the **Municipal Officer Triage Panel**. Drainage relief teams deploy before widespread flooding paralyzes the city corridor.

2. Full-Stack Technical Architecture

Component / Layer	Technology Specification	Core Architectural Purpose
Frontend Framework	Next.js 16 + React 19 (App Router)	High-performance server-rendered hydration, live state reactivity, mobile-first UX.
Styling System	Tailwind CSS v4 + Lucide Icons	Adaptive dark/light glassmorphic UI, responsive incident status badges, animations.
Database Storage	MongoDB 7.0 + Mongoose 9.1	2dsphere GeoJSON spatial querying, raw moment storage, and automatic TTL index expiry.
Multimodal AI Vision	Groq API (Llama 3.2 11B Vision)	Sub-second multimodal vision inference: flood depth, OCR, and anti-spam verification.
Media Pipeline	Cloudinary v2 SDK	Direct Base64 media ingestion, optimized CDN hosting, automatic responsive resizing.
Meteo Ingestion Suite	Open-Meteo, WeatherAPI, OpenWeather	Triple-vendor parallel API dispatch, normalization, consensus aggregation.

3. Core Algorithmic Systems & Mathematical Scoring

A. Triple-Provider Consensus Scoring Engine (aggregator.ts)

Commercial weather apps fail when their single API source goes stale or drifts. WeatherPulse resolves this by fetching data in parallel from three heterogeneous APIs (Open-Meteo, WeatherAPI, OpenWeather) and applying a composite reliability formula yielding a confidence score **S ∈ [0, 100]**:

```
Consensus Formula: S = 0.40 · S_agreement + 0.30 · S_freshness + 0.30 · S_availability
```

- **Agreement Score (S_agreement):** Compares temperature variance (ΔT) and condition taxonomy across vendors. Temperature dispersion: *Score_temp = max(0, 100 - (ΔT_max · 20))*. A deviation > 5°C drops score to 0. Condition matching computes intersection between rain/storm versus dry conditions. Composite: *0.6 · Score_temp + 0.4 · Score_condition*.
- **Freshness Score (S_freshness):** Continuous temporal decay function based on timestamp age: *Score_fresh = max(0, min(100, 100 - (Age_minutes / 60) · 100))*. Readings < 10 mins old receive 100%; readings older than 60 mins decay to zero.
- **Availability Score (S_availability):** Ratio of healthy HTTP 200 responses against total configured provider endpoints (N_ok / N_total · 100).
- **Disagreement Flagging:** If temperature delta exceeds 3.0°C or condition codes collide (e.g., Rain vs Clear), automated warning flags are injected for user transparency.

B. Spatial-Temporal Haversine Incident Clustering (verificationEngine.ts)

When severe weather strikes, multiple citizens report the same incident. Without intelligent deduplication, emergency personnel are overwhelmed. The clustering engine executes the Great-Circle Haversine distance algorithm:

```
a = sin²(Δϕ / 2) + cos(ϕ1) · cos(ϕ2) · sin²(Δλ / 2)
d = 2 · R · atan2(√a, √(1 - a)) where R = 6,371,000 meters (Earth's radius)
Clustering Constraint: If distance d ≤ 500m AND |t1 - t2| ≤ 30 mins → Corroborate into single IncidentCluster.
```

- **Corroboration Confidence Boost:** An isolated report starts as 'unverified' (Confidence: 50%). If 2 or more citizens report within 500m / 30 mins, status upgrades to 'corroborated' (+25%) and 'confirmed' (+40%).
- **Physical Plausibility Check:** Cross-checks cluster reports against current Doppler precipitation. If rain is reported when radar indicates zero cloud cover, report priority is demoted to prevent malicious spoofing.

4. Multimodal AI Vision Engine (Groq & Llama 3.2 Vision)

Uploaded citizen photographs are processed through the ultra-fast **Llama 3.2 11B Vision** model via Groq LPUs (**groqVision.ts**). The vision engine returns structured JSON evaluation covering 5 critical metrics:

Vision Metric	Algorithmic Function & Output Schema
1. Authenticity Screening	Detects and blocks invalid uploads (selfies, indoor rooms, computer monitors, internet memes, food). Emits isValidWeatherPhoto: boolean.
2. Flood Depth Estimation	Computes surface water reflection, tire submergence, and curb visibility. Categorizes depth into: none | ankle_deep | knee_deep | submerged_vehicles.
3. Zero-Shot Signboard OCR	Extracts optical text from commercial boards, mileposts, or street nameplates to independently confirm user GPS location.
4. Hazard Classification	Identifies secondary threats such as fallen electric wires, low-visibility downpours, or open drains, producing detectedHazards: string[].
5. Executive Triage Summary	Generates an actionable 2-sentence English brief for municipal disaster officers with severity: CRITICAL, HIGH, MEDIUM, or LOW.

5. UI/UX Design Process & Engineering Workflows

- **Camera Subsystem (CameraCapture.tsx):** Direct HTML5 media stream with live canvas snapshot capture and system file-picker fallback.
- **High-Accuracy Geolocation:** Dual-tier GPS lock with timeout failsafe, integrated with predefined coordinate matrices for tier-1/tier-2 Indian cities.
- **Officer Triage Panel (OfficerTriagePanel.tsx):** Live administrative incident board supporting severity sorting, coordinate review, and 1-click status transitions.
- **Ephemeral Data Retention:** Database records utilize MongoDB TTL (Time-To-Live) indexing to expire raw community moments automatically after 4–12 hours, ensuring minimal storage footprint.

6. Architectural Summary & Strategic Verdict

Strategic Verdict: *WeatherPulse India transforms meteorological observation from passive remote estimation into a verified, real-time civic defense network. By uniting triple-provider consensus scoring, Haversine spatial-temporal clustering, and Groq Llama 3.2 Vision intelligence, the platform equips city administrators and citizens with unparalleled hyper-local disaster response capabilities.*